

AI Agents - Building Intelligent Automation

From Concept to Working Agent

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Algorithm Basics and BW

What We'll Cover Today

1.  What are AI agents?
2.  Agent architectures
3.  Tools for building agents
4.  Workflow automation
5.  Live demo: File renaming agent
6.  Hands-on: Build your own agent

Quick Recap: AI in Business

Last class we covered:

- ▶  Finance AI applications
- ▶  Operations optimization
- ▶  Accounting automation
- ▶  AI strategy & implementation
- ▶  ROI measurement
- ▶  Group project proposals

Today: Build AI agents that automate workflows autonomously

What Are AI Agents? Autonomous systems that perceive, decide, and act

Key characteristics:

- ▶  **Perception:** Sense environment
 - ▶ Read data, monitor events
- ▶  **Reasoning:** Process information
 - ▶ Understand context, make decisions
- ▶  **Action:** Execute tasks
 - ▶ Perform operations autonomously
- ▶  **Learning:** Improve over time
 - ▶ Adapt from feedback

Simple analogy:

- ▶  **Human assistant:**
 - ▶ Checks email (perceive)
 - ▶ Decides priority (reason)
 - ▶ Responds or flags (act)
 - ▶ Learns preferences (improve)
- ▶  **AI agent:**
 - ▶ Same pattern, automated!

Key difference from tools: Agents act independently, tools wait for commands

AI Tools vs AI Agents

Aspect	AI Tools	AI Agents
User interaction	Passive, waits for input	Proactive, acts autonomously
Decision-making	Follows exact instructions	Makes independent decisions
Workflow	Single task per request	Multi-step workflows
Adaptability	Static responses	Learns and adapts
Examples	ChatGPT, DALL-E	Scheduling bots, monitoring systems
Control	Full human control	Delegated authority

Think: Tool = hammer, Agent = construction worker with hammer

Types of AI Agents

By complexity:

1.  **Simple reflex:**
 - ▶ If-then rules
 - ▶ No memory
 - ▶ Example: Thermostat
2.  **Model-based:**
 - ▶ Maintains state
 - ▶ Considers history
 - ▶ Example: Chess AI
3.  **Goal-based:**
 - ▶ Plans to achieve goals
 - ▶ Example: GPS navigation
4.  **Utility-based:**
 - ▶ Optimizes outcomes
 - ▶ Example: Stock trading bot
5.  **Learning agents:**
 - ▶ Improve from experience
 - ▶ Example: Recommendation systems

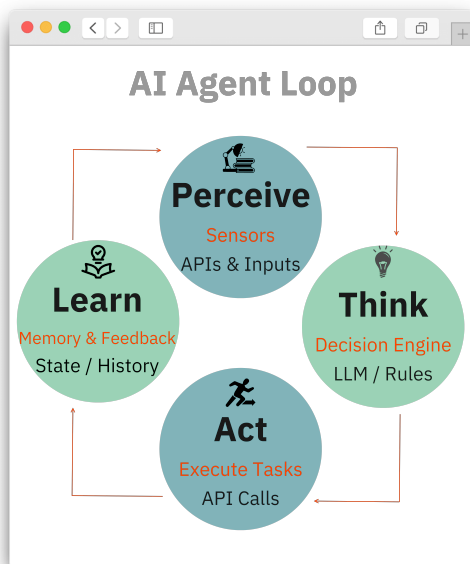
Today's focus:

Goal-based agents with learning

Agent Architecture Components

Core components:

1. **👁️ Sensors:**
 - ▶ APIs, webhooks, file watchers
2. **🧠 Decision engine:**
 - ▶ LLM, rules, ML models
3. **⚙️ Actuators:**
 - ▶ API calls, file operations
4. **💾 Memory:**
 - ▶ State, history, context



Real-World AI Agent Examples

Customer service:

- ▶  Chatbots that resolve issues
- ▶  Ticket routing & escalation
- ▶  Voice assistants
- ▶  Email auto-responders

Business operations:

- ▶  Meeting schedulers
- ▶  Invoice processors
- ▶  Report generators
- ▶  Alert monitors

Personal productivity:

- ▶  Email sorters (SaneBox)
- ▶  Calendar optimizers
- ▶  Shopping assistants
- ▶  News curators

Advanced examples:

- ▶  Autonomous vehicles
- ▶  Game NPCs
- ▶  Trading bots
- ▶  Smart home systems

Why Build AI Agents?

Business benefits:

- ▶  24/7 operation
 - ▶ No downtime
 - ▶ Instant response
- ▶  Scalability
 - ▶ Handle volume spikes
 - ▶ No hiring needed
- ▶  Cost efficiency
 - ▶ 60-80% cost reduction
 - ▶ One-time setup

Operational advantages:

- ▶  Consistency
 - ▶ No human error
 - ▶ Same quality always
- ▶  Speed
 - ▶ Instant execution
 - ▶ Parallel processing
- ▶  Employee satisfaction
 - ▶ Eliminate boring tasks
 - ▶ Focus on creative work

Agent Workflow Patterns: Common automation patterns

1. **Trigger-Action:**

- ▶ Event occurs → Agent acts
- ▶ Example: New email → Categorize & respond

2. **Scheduled:**

- ▶ Time-based execution
- ▶ Example: Daily report generation at 9am

3. **Monitoring:**

- ▶ Continuous observation → Alert on condition
- ▶ Example: Website down → Notify team

4. **Data transformation:**

- ▶ Input → Process → Output
- ▶ Example: CSV upload → Clean → Database

Tools for Building AI Agents

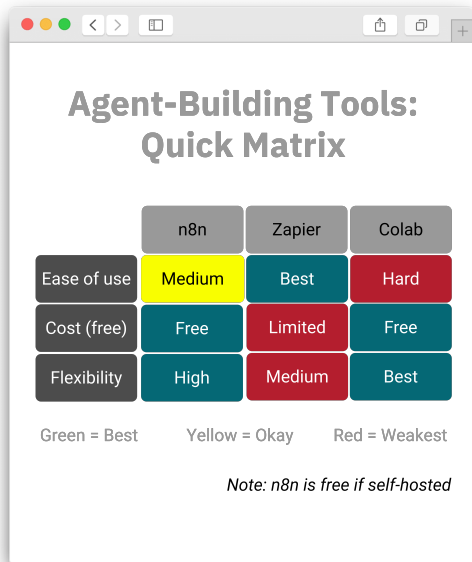
No-code/Low-code:

- ▶  **n8n.io**
 - ▶ Visual workflow builder
 - ▶ 400+ integrations
 - ▶ Free self-hosted
- ▶  **Zapier**
 - ▶ Easy automation
 - ▶ 5000+ apps
 - ▶ Limited free tier
- ▶  **Make.com**
 - ▶ Visual scenarios
 - ▶ Complex workflows
 - ▶ Free tier available

Code-based:

- ▶  **Python**
 - ▶ LangChain, AutoGPT
 - ▶ Full flexibility
 - ▶ Coding required
- ▶  **JavaScript**
 - ▶ Node.js workflows
 - ▶ Web integration
- ▶  **Google Colab**
 - ▶ Free Python notebooks
 - ▶ GPU access
 - ▶ Good for learning

Tool Comparison: n8n vs Zapier vs Colab



Agent-Building Tools: Quick Matrix

	n8n	Zapier	Colab
Ease of use	Medium	Best	Hard
Cost (free)	Free	Limited	Free
Flexibility	High	Medium	Best

Green = Best Yellow = Okay Red = Weakest

Note: n8n is free if self-hosted

Quick comparison:

▶ **n8n:**

- ▶ Best for complex workflows
- ▶ Free (self-hosted)
- ▶ Learning curve: Medium

▶ **Zapier:**

- ▶ Easiest to start
- ▶ \$\$ (limited free)
- ▶ Learning curve: Low

▶ **Colab:**

- ▶ Most flexible
- ▶ Free forever
- ▶ Learning curve: High (coding)

Single-Agent vs Multi-Agent Systems

Single-agent:

- ▶  One agent, one task
- ▶  Linear workflow
- ▶  Simpler to build
- ▶  Easier to debug

Example:

- ▶ Email categorizer
- ▶ File organizer
- ▶ Report generator

Today's focus: Single-agent (perfect for learning!)

Multi-agent:

- ▶  Multiple specialized agents
- ▶  Parallel workflows
- ▶  More powerful
- ▶  Complex coordination

Example:

- ▶ Customer service system:
 - ▶ Routing agent
 - ▶ Resolution agent
 - ▶ Escalation agent

Agent Communication & APIs: How agents connect to the world

Input methods:

- ▶  **APIs:**
 - ▶ REST, GraphQL
 - ▶ Read data from services
- ▶  **File watchers:**
 - ▶ Monitor directories
 - ▶ Trigger on new files
- ▶  **Webhooks:**
 - ▶ Push notifications
 - ▶ Event-driven

Output methods:

- ▶  **API calls:**
 - ▶ Write to services
 - ▶ Update databases
- ▶  **File operations:**
 - ▶ Create, modify, move
- ▶  **Notifications:**
 - ▶ Email, Slack, SMS

Key: APIs are the “nervous system” connecting agents to tools

LLMs as Agent Brains: Why LLMs power modern agents

Capabilities:

- ▶  Natural language understanding
- ▶  Context reasoning
- ▶  Decision-making
- ▶  Tool use (function calling)
- ▶  Knowledge retrieval

Agent workflow with LLM:

1. Input: “New email from client”
2. LLM analyzes: Urgent? Priority?
3. LLM decides: Respond or escalate?
4. LLM generates: Draft response
5. Agent executes: Send email

Popular LLM APIs: OpenAI (GPT-4), Anthropic (Claude), Google (Gemini)

Today's Project: File Renaming Agent: What we'll build

Problem:

- ▶  Messy filenames
 - ▶ "IMG_1234.jpg"
 - ▶ "document.pdf"
 - ▶ "scan0001.png"
- ▶  Hard to find files
- ▶  Waste time organizing

Example transformation:

- ▶ "IMG_1234.jpg" → "2024_Q4_Sales_Chart.jpg"
- ▶ "document.pdf" → "Invoice_Acme_Corp_Dec2024.pdf"

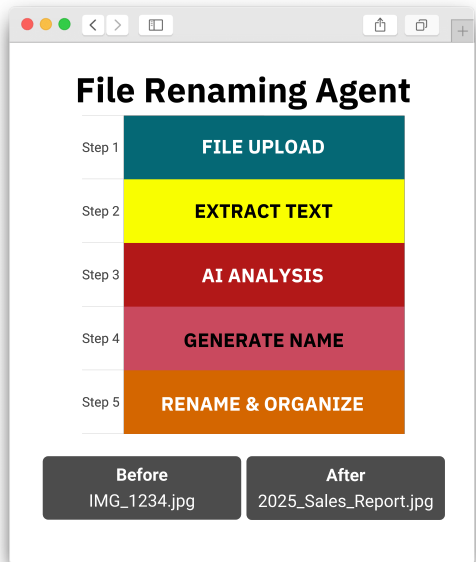
Solution:

- ▶  AI agent analyzes content
- ▶  Generates descriptive name
- ▶  Automatically renames
- ▶  Organized files!

File Renaming Agent Architecture

Agent workflow:

1. 📁 User uploads file
2. 👁 Agent reads content
 - ▶ OCR for images
 - ▶ Text extraction
3. 🧠 LLM analyzes content
 - ▶ Understand topic
 - ▶ Generate name
4. ✏️ Rename file
5. ✅ Move to organized folder



Technical Components We'll Use

Platform:

- ▶  Google Colab
 - ▶ Free Python environment
 - ▶ No installation needed
 - ▶ Works in browser

Libraries:

- ▶  Python 3
- ▶  OpenAI API (for LLM)
- ▶  File handling (built-in)
- ▶  PIL (image processing)

What you need:

- ▶  Google account
 - ▶ For Colab access
- ▶  OpenAI API key (optional)
 - ▶ Free tier: \$5 credit
 - ▶ Or use provided key
- ▶  Web browser
- ▶  Sample files to rename

No coding experience needed! Just click "Run" on each cell

Getting OpenAI API Key: If you want your own key

Steps:

1.  Go to platform.openai.com
2.  Sign up (free)
3.  API Keys → Create new
4.  Copy the key
5.  \$ \$5 free credit included

Alternative:

- ▶  Use instructor-provided key
 - ▶ Shared for class
 - ▶ Limited usage
- ▶  Don't share your key
- ▶  \$ Monitor usage

Today: Instructor provides key, so you can start immediately!

Live Demo Time!

Follow along on your laptops

Demo Step 1: Access Colab Notebook

Follow these steps:

1.  Open browser
2.  Go to:
<https://bit.ly/ai-file-agent>
3.  Sign in with Google account
4.  Click "Copy to Drive"
 - ▶ Makes your own editable copy
5.  Ready to run!

Notebook structure:

- ▶ Cell 1: Setup & imports
- ▶ Cell 2: Upload files
- ▶ Cell 3: AI processes & renames
- ▶ Cell 4: Download results

Demo Step 2: Cell 1 - Setup

What this cell does:

- ▶  Install required libraries
- ▶  Configure API key
- ▶  Verify connection

You'll see:

- ▶ "Installing dependencies..."
- ▶ "Setup complete!"

Action:

1. Click the Play button on Cell 1
2. Wait for completion (10-15 seconds)
3. Look for green checkmark

Demo Step 3: Cell 2 - Upload Files

What this cell does:

- ▶  Shows file upload button
- ▶  Accepts any file type
- ▶  Lists uploaded files

Action:

1. Click Play on Cell 2
2. Click "Choose Files" button
3. Select 3-5 files with messy names and click "Open"

Good test files:

- ▶ Screenshots, photos, scanned documents, PDFs

Demo Step 4: Cell 3 - AI Processes Files

What this cell does:

- ▶  Reads each file's content
- ▶  Sends to AI for analysis
- ▶  AI generates descriptive name
- ▶  Renames file
- ▶  Shows before/after

Action:

1. Click Play on Cell 3
2. Watch the magic happen!
3. See progress for each file
4. Review before/after names

Takes: 5-10 seconds per file

Demo Step 5: Cell 4 - Download Results

What this cell does:

- ▶  Zips renamed files
- ▶  Creates download link
- ▶  Save to your computer

Compare:

- ▶ Before: "IMG_1234.jpg"
- ▶ After:
"2024_Team_Meeting_Photo.jpg"

Action:

1. Click Play on Cell 4
2. Click "Download" link
3. Extract zip file
4. See beautifully renamed files!

Understanding the Code: What's happening behind the scenes

File reading:

- ▶  Python reads file
- ▶  Images: OCR extraction
- ▶  PDFs: Text extraction
- ▶  Docs: Content parsing

AI prompt:

- ▶ "Analyze this content"
- ▶ "Generate descriptive filename"
- ▶ "Format:
category_description_date"

Key: You don't write code - just configure & run!

AI response:

- ▶  AI understands content
- ▶  Generates appropriate name
- ▶  Returns structured format

File operations:

- ▶  Rename file
- ▶  Organize by type
- ▶  Save to output folder

Hands-On: Now you build it!

1.  Open your copy of the notebook
2.  Upload your own messy files
3.  Run all cells
4.  Watch the agent work
5.  Download renamed files
6.  Enjoy organized files!

Real-World Agent Applications: Similar agents businesses use

Document processing:

- ▶  Invoice extraction
- ▶  Contract analysis
- ▶  Receipt categorization
- ▶  ID verification

Customer service:

- ▶  Email auto-responder
- ▶  Chat support
- ▶  Call transcription
- ▶  Ticket routing

Operations:

- ▶  Report generation
- ▶  Data entry automation
- ▶  System integration
- ▶  Alert monitoring

Market size:

- ▶ RPA market: \$10B+ (2024)
- ▶ AI agents: Growing 40%/year
- ▶ Every company needs this

Building More Complex Agents: Next-level agent capabilities

1. **Multi-step workflows:**

- ▶ Chain multiple actions
- ▶ Conditional logic
- ▶ Error handling

2. **State management:**

- ▶ Remember past actions
- ▶ Learn preferences
- ▶ Track progress

3. **Human-in-the-loop:**

- ▶ Request approval for critical actions
- ▶ Provide feedback
- ▶ Override decisions

Agent Development Best Practices

Design principles:

- ▶  Start simple, iterate
- ▶  Test thoroughly
- ▶  Handle errors gracefully
- ▶  Design for users
- ▶  Document behavior

Security:

- ▶  Secure API keys
- ▶  Limit permissions
- ▶  Validate inputs
- ▶  Audit logs

Monitoring:

- ▶  Track performance
- ▶  Monitor errors
- ▶  Watch costs
- ▶  User feedback

Maintenance:

- ▶  Update models
- ▶  Fix issues promptly
- ▶  Improve continuously
- ▶  Version control

Ethical Considerations for Agents

Key concerns:

- ▶  **Bias:**
 - ▶ AI reflects training data
 - ▶ Can perpetuate unfairness
 - ▶ Test diverse scenarios
- ▶  **Transparency:**
 - ▶ Users should know it's AI
 - ▶ Provide override options
- ▶  **Privacy:**
 - ▶ Protect user data
 - ▶ Comply with regulations
 - ▶ Secure storage

Responsible deployment:

- ▶  Human oversight
 - ▶ Critical decisions reviewed
- ▶  Fail-safe mechanisms
 - ▶ Stop if uncertain
- ▶  User control
 - ▶ Easy to pause/stop
- ▶  Accountability
 - ▶ Clear responsibility

Rule: Agents augment humans, don't replace judgment

Career Opportunities in Agent Development

Roles:

- ▶  AI Agent Developer
 - ▶ \$90k-180k
 - ▶ Build automation
- ▶  Automation Engineer
 - ▶ \$80k-150k
 - ▶ RPA + AI integration
- ▶  AI Kill Switch Operator
 - ▶ \$300k-500k
 - ▶ Kill AI if sentient

Skills needed:

- ▶  Python programming
- ▶  LLM integration
- ▶  API knowledge
- ▶  Workflow design
- ▶  Problem-solving

Good news:

- ▶ High demand
- ▶ Entry-level friendly
- ▶ Rapid skill growth

Key Takeaways

1. 🤖 AI agents = Autonomous systems (perceive, reason, act)
2. 🧠 LLMs enable intelligent decision-making
3. ⚙️ No-code tools (n8n, Zapier) make agents accessible
4. </> Code-based (Python/Colab) offers flexibility
5. 📄 You built a working agent today!
6. 📦 Real business value: 60-80% efficiency gains
7. 🚀 Foundation for complex automation

Remember: Start simple, solve real problems, iterate based on feedback

Questions?